

May-25-0748

MAFC-311 (Probability Theory and Statistics)

B.Tech 3rd-4th-6th (NEP) CS & E

Time : 3 Hours

Max. Marks : 60

Note : The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, one question each from sections A, B, C, D and Section E is compulsory. Use of non-programmable calculator is allowed.

SECTION - A

1. (a) The contents of urns I, II and III are as follows: Urn I: 1 white, 2 black and 3 red balls; Urn II: 2 white, 1 black and 1 red balls; and Urn III: 4 white, 5 black and 3 red balls. One urn is chosen at random and two balls drawn. They happen to be white and red. What is the probability that they came from urns I, II or III? **(6)**

(b) The joint pdf of a two-dimensional random variable (X, Y) is $f(x,y) = 2$, for $0 < x < 1$, $0 < y < x$, and 0 elsewhere. Find the marginal density functions of X and Y . **(6)**

OR

2. The mean and variance of a binomial distribution are 4 and (a given value) respectively. Find $P(X \geq 1)$. **(6)**

SECTION - B

3. (a) In a large lot of electric bulbs, the mean life and standard deviation are 2160 hours and 90 hours respectively. A sample of 100 bulbs is chosen. It is found that the mean life and standard deviation of the bulbs in the sample are 2155 hours and 90 hours respectively. Can we conclude that the sample is drawn from the given population? If we assume that the population is normal and its mean is unknown, find the 95% confidence limits of the mean (given that the critical value z_{α} of Z for a two-tailed test at 5% level of significance is 1.96). **(6)**

(b) In a city A, 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant? **(6)**

OR

4. (a) In two large populations there are 30% and 25% respectively of fair-haired people. Is this difference likely to be hidden in samples of 1200 and 900 respectively drawn from the two populations? **(6)**

(b) The mean breaking strength of cables supplied by a manufacturer is 1800 with a standard deviation of 100. By a new technique in the manufacturing process it is claimed that the breaking strength of the cables has increased. In order to test this claim, a sample of 50 cables is tested and the mean breaking strength is found to be 1850. Can we support the claim at 0.01 level of significance? (given that the critical value z_{α} for a right-tailed test at 1% level of significance is 2.33). **(6)**

SECTION - C

5. (a) Using three iterations of the Newton-Raphson method, obtain approximate values of $\sqrt{11}$ and $1/17$. Use a suitable initial approximation. **(6)**

(b) Given the values: $x = 0, 1, 3, 4, 7$ and $f(x) = 1, 3, 49, 129, 813$, evaluate $f(0.3)$ using Newton's divided difference formula. **(6)**

OR

6. (a) Find the Lagrange interpolating polynomial that fits the following data values: $x = -1, 2, 3, 4$ and $f(x) = -1, 11, 31, 69$. Determine the approximate value of $f(1.5)$. **(6)**
- (b) A root of the equation $xe^x - 1 = 0$ lies in the interval $(0.5, 1)$. Determine this root correct to three decimal places using the regula-falsi method. **(6)**

SECTION - D

7. (a) Using Simpson's 3/8th rule, evaluate the following integral with 4 nodal points: $\int_1^3 \frac{x}{(x^2+2x+2)} dx$ over $[1, 3]$. **(6)**
- (b) Given that $dy/dx = x^2 + y$ and $y(0) = 1$, find an approximate value of $y(0.1)$ taking $h = 0.05$ by the modified Euler's method. **(6)**

OR

8. (a) Solve the initial value problem $dy/dx = \sqrt{x+y}$, $y(0.4) = 0.41$, in the interval $[0.4, 0.8]$ using the Runge-Kutta fourth order method with step size $h = 0.2$. **(6)**
- (b) Obtain the approximate value of $y(1.3)$ for the initial value problem $dy/dx = -2xy^2$, $y(1) = 1$ using (i) Taylor series (ii) second order method, with step size $h = 0.1$. **(12)**

SECTION - E

(Compulsory)

9. (a) A continuous random variable x has the pdf $f(x) = k/x^2$ for $15 < x < 25$ and 0 elsewhere, where k is a constant. Determine the constant k .
- (b) A continuous random variable x has the pdf $f(x) = A + Bx$ for $0 \leq x \leq 1$. If the mean of the distribution is $2/3$, find A and B .
- (c) Find the second divided difference of $f(x) = 1/x$ using the points x_0, x_1, x_2 .
- (d) The number of students in a class is 100. The average marks scored by 54 boys is 60 with standard deviation 8, while the average marks scored by 46 girls is 73 with standard deviation 8. Test at 1% level of significance whether the girls performed better than the boys (given critical value $z = 2.33$ for a right-tailed test at 1% level of significance).
- (e) Evaluate $\int_1^3 \frac{dx}{(3+2x)}$ over $[1, 3]$ using Simpson's 1/3rd rule with two intervals.
- (f) Using Picard's method, solve $dy/dx = -xy$ with $x_0 = 0$, $y_0 = 1$, up to two approximations. **(6×2=12)**

Transcribed from a photograph of the original answer/question booklet. A few numeric values, table entries and one section could not be fully verified due to image quality (angle/blur/overlap with adjacent papers) — please cross-check against your original copy.